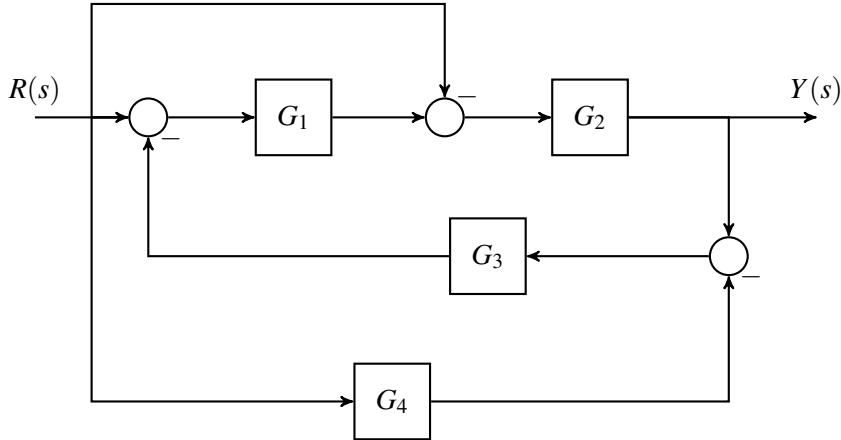


ECE 4375, Exam #1 Review

1. Given: The block diagram

Find: The transfer function, $\frac{Y(s)}{R(s)}$



2. Given: A series of mixed domain equations are used to describe a process, The input to the system is r , the overall output is y . All other signals are internal to the system; assume there are no initial conditions.

$$E(s) = R(s) - \frac{9}{s+9}Y(s) \quad (1a)$$

$$U(s) = \frac{4}{s}E(s) \quad (1b)$$

$$V(s) = \frac{3}{s+3}U(s) \quad (1c)$$

$$\ddot{x}(t) + 4\dot{x}(t) + 5x(t) - \dot{v}(t) - v(t) = 0 \quad (1d)$$

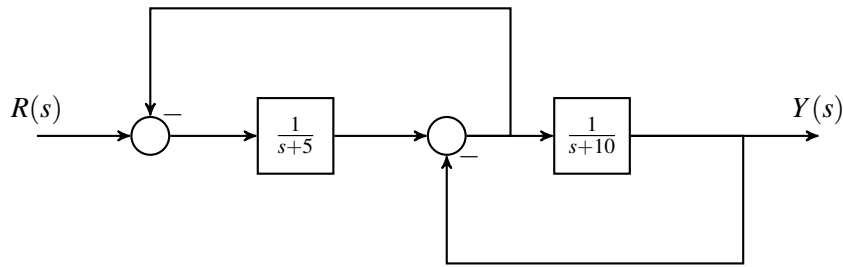
$$y(t) - 7x(t) = 0. \quad (1e)$$

Find:

A. The open-loop transfer function, $\frac{Y(s)}{R(s)}$

B. From A find the loop-gain

3. Given the block diagram in



Find: Find:

- A. The open-loop transfer function, $\frac{Y(s)}{R(s)}$
- B. From A find the loop-gain

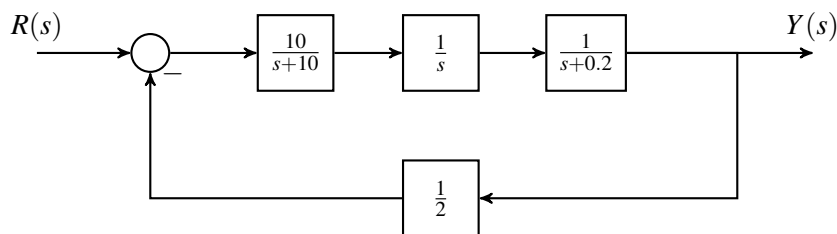
4. Given:

$$V(s) = \frac{400}{s^2 + 8s + 400} \quad (2)$$

Find:

- A. $v(t)$
- B. $v(0)$ using
 1. IVT
 2. evaluating the time domain equation when $t = 0$
- C. $v(\infty)$ using
 1. FVT
 2. evaluating the time domain equation when $t \rightarrow \infty$
- D. $v(t = 1)$

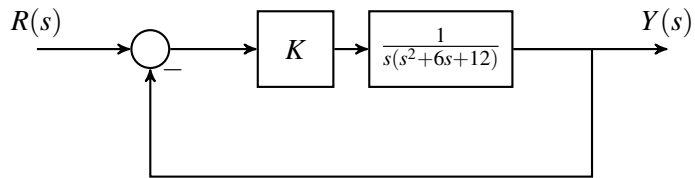
5. Given the block diagram



Find:

- A. The transfer function, $\frac{Y(s)}{R(s)}$
- B. $y(0)$ using IVT when the input is an i) impulse and ii) a unit step
- C. $v(\infty)$ using FVT when the input is an i) impulse and ii) a unit step (determine stability with the Routh Array)

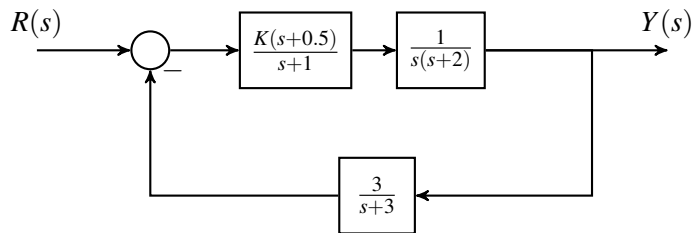
6. Given the block diagram



Find:

- A. The transfer function, $\frac{Y(s)}{R(s)}$
- B. The loop-gain

7. Given the block diagram



Find:

- A. The transfer function, $\frac{Y(s)}{R(s)}$
- B. The loop-gain

8. Given: the characteristic polynomial:

$$A(s) = s^3 + 8s^2 + 29s + 52 \quad (3)$$

Find:

- A. The number of unstable roots

9. Given: the characteristic polynomial:

$$A(s) = s^3 - 3s - 52 \quad (4)$$

Find:

- A. The number of unstable roots

10. Given: the characteristic polynomial:

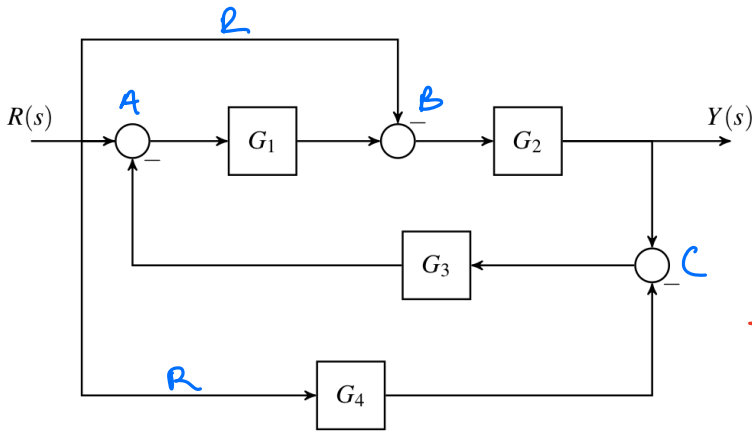
$$A(s) = s^3 + 4s^2 + 13s \quad (5)$$

Find:

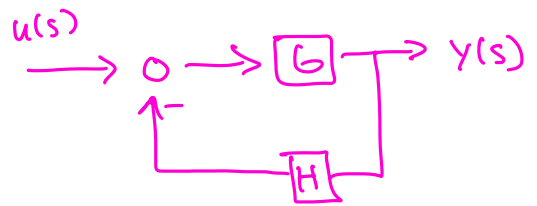
- A. The number of unstable roots

1. Given: The block diagram

Find: The transfer function, $\frac{Y(s)}{R(s)}$



Recall:



$$\frac{y(s)}{u(s)} = \frac{G}{1+GH}$$

$\Rightarrow$ Use node eqs. when block reduction isn't straight fwd.

$\Rightarrow$ Summation Junctions can be labeled as nodes

$$A = R - G_3 C$$

$$B = A G_1 - R$$

$$C = Y - G_4 R$$

$$Y = B G_2$$

$$Y = (A G_1 - R) G_2$$

$$Y = [(R - G_3 C) G_1 - R] G_2$$

$$Y = [R G_1 - G_1 G_3 C - R] G_2$$

$$Y = [R G_1 - G_1 G_3 (Y - G_4 R) - R] G_2$$

$$Y = [R G_1 - G_1 G_3 Y + G_1 G_3 G_4 R - R] G_2$$

$$Y = R G_1 G_2 - G_1 G_2 G_3 Y + G_1 G_2 G_3 G_4 R - R G_2$$

$$Y + G_1 G_2 G_3 Y = R G_1 G_2 + G_1 G_2 G_3 G_4 R - R G_2$$

$$Y(1 + G_1 G_2 G_3) = R(G_1 G_2 + G_1 G_2 G_3 G_4 - G_2)$$

$$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 + G_1 G_2 G_3 G_4 - G_2}{1 + G_1 G_2 G_3}$$

2. Given: A series of mixed domain equations are used to describe a process, The input to the system is r , the overall output is y . All other signals are internal to the system; assume there are no initial conditions.

$$E(s) = R(s) - \frac{9}{s+9}Y(s) \quad (1a)$$

$$U(s) = \frac{4}{s}E(s) \quad (1b)$$

$$V(s) = \frac{3}{s+3}U(s) \quad (1c)$$

$$\ddot{x}(t) + 4\dot{x}(t) + 5x(t) - \dot{v}(t) - v(t) = 0 \quad (1d)$$

$$y(t) - 7x(t) = 0. \quad (1e)$$

Transform to Laplace domain

Find:

A. The open-loop transfer function, $\frac{Y(s)}{R(s)}$

B. From A find the loop-gain

$R(s) \rightarrow$ intermediate transfer functions $\rightarrow Y(s)$
 * Goal is to narrow equation down in terms of $Y(s)$ and $R(s)$

(a)
 eq 1d] $s^2 X(s) + 4sX(s) + 5X(s) - sV(s) - V(s) = 0$

$$X(s)(s^2 + 4s + 5) = V(s)(s + 1)$$

$$X(s) = V(s) \left(\frac{s+1}{s^2 + 4s + 5} \right)$$

eq 1e] $V(s) - 7X(s) = 0$

$$V(s) = 7X(s)$$

$$V(s) = 7V(s) \left(\frac{s+1}{s^2 + 4s + 5} \right) \Rightarrow V(s) = \left(\frac{V(s)}{7} \right) \left(\frac{s^2 + 4s + 5}{s+1} \right)$$

eq 1b] $U(s) = \frac{4}{s}E(s)$

$$E(s) = \frac{s}{4}U(s)$$

eq 1c] $V(s) = \frac{3}{s+3}U(s)$

$$U(s) = \frac{s+3}{3}V(s)$$

$$\Rightarrow E(s) = \frac{s}{4} \left(\frac{s+3}{3} v(s) \right) = \frac{s(s+3)}{12} v(s)$$

$$E(s) = \left(\frac{s(s+3)}{12} \right) \left(\frac{v(s)}{7} \right) \left(\frac{s^2+4s+5}{s+1} \right)$$

$$\Rightarrow v(s) \left(\frac{s(s+3)(s^2+4s+5)}{84(s+1)} \right) = R(s) - \frac{9}{s+9} v(s)$$

∴ don't plug in what $v(s)$ is, solve for $\frac{v(s)}{R(s)}$

$$v(s) \left(\frac{s(s+3)(s^2+4s+5)}{84(s+1)} \right) + \frac{9}{s+9} v(s) = R(s)$$

$$v(s) \left[\left(\frac{s(s+3)(s^2+4s+5)}{84(s+1)} \right) + \frac{9}{s+9} \right] = R(s)$$

$$v(s) \left[\frac{(s^2+3s)(s^2+4s+5)(s+9)}{84(s+1)(s+9)} + \frac{9(84)(s+1)}{84(s+1)(s+9)} \right] = R(s)$$

$$v(s) \left[\frac{s^5 + 16s^4 + 80s^3 + 168s^2 + 135s + 756s + 756}{84s^2 + 840s + 756} \right] = R(s)$$

$$\frac{v(s)}{R(s)} = \left[\frac{84s^2 + 840s + 756}{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756} \right]$$

(b) Loop Gain, GH , $L(s)$

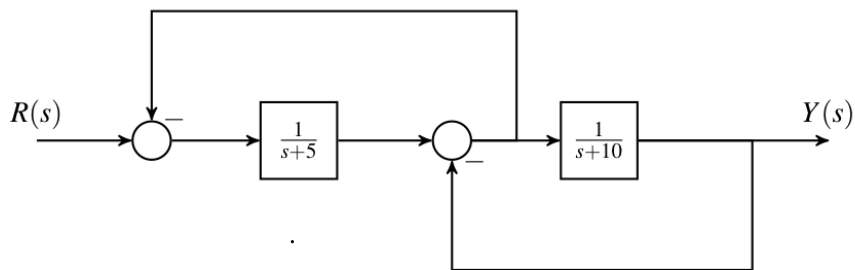
$$\frac{Y(s)}{R(s)} = \frac{\frac{84s^2 + 840s + 756}{84s^2 + 840s + 756}}{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756} = \frac{\frac{6}{6}}{\frac{1 + 6H}{6}}$$

$$= \frac{1}{\frac{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756}{84s^2 + 840s + 756} + \frac{756}{84s^2 + 840s + 756}} = \frac{1}{\frac{1}{6} + H}$$

$$G = \frac{84s^2 + 840s + 756}{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756}, \quad H = \frac{756}{84s^2 + 840s + 756}$$

$$L(s) = GH = \frac{84s^2 + 840s + 756}{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756} \cdot \frac{756}{84s^2 + 840s + 756} = \boxed{\frac{756}{s^5 + 16s^4 + 80s^3 + 168s^2 + 891s + 756}}$$

3. Given the block diagram in



Find: Find:

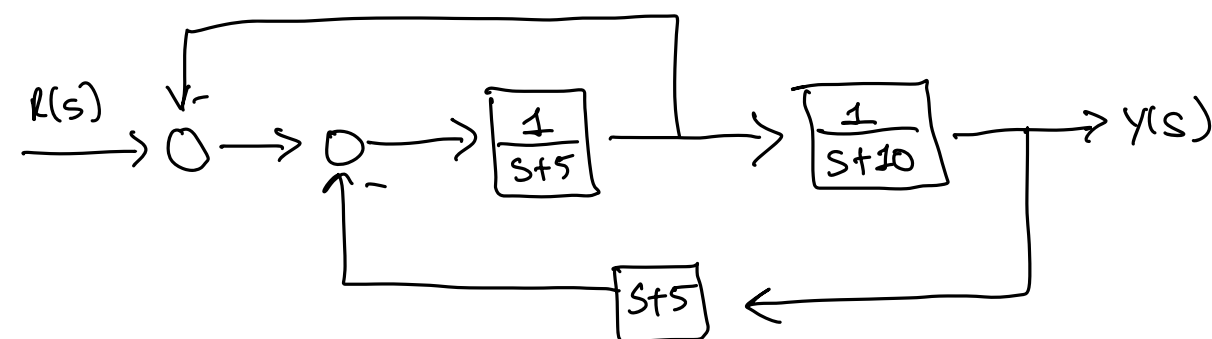
A. The open-loop transfer function, $\frac{Y(s)}{R(s)}$

B. From A find the loop-gain

(A)

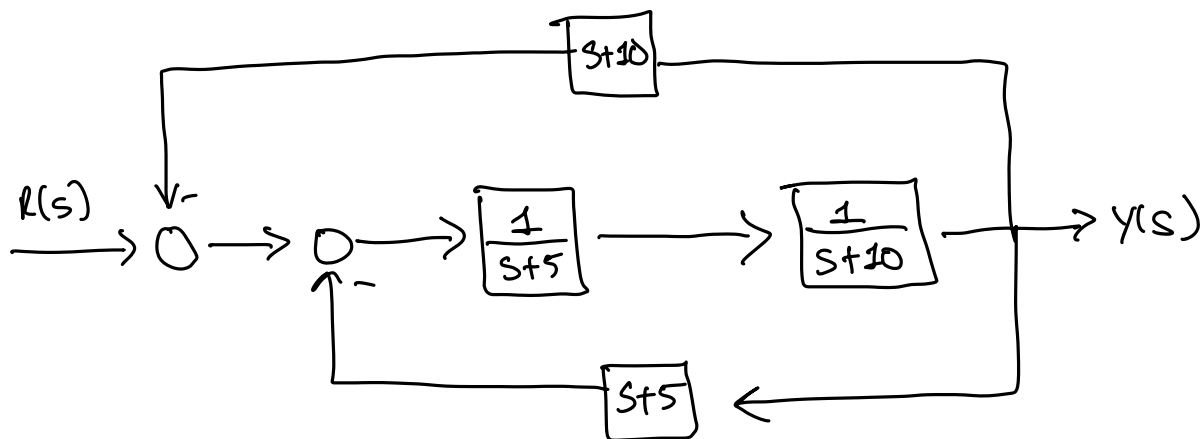
Step 1: Move summation junction before $\boxed{\frac{1}{s+5}}$

$$\frac{1}{\frac{1}{s+5}} = s+5$$



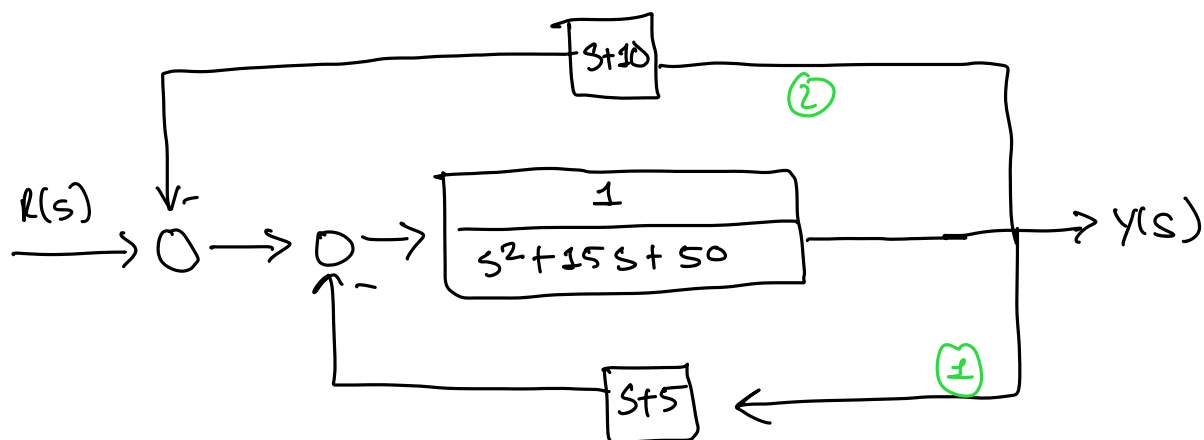
Step 2: Move Branch after $\boxed{\frac{1}{s+10}}$

$$\frac{\frac{1}{s+10}}{\frac{1}{s+10}} = s+10$$



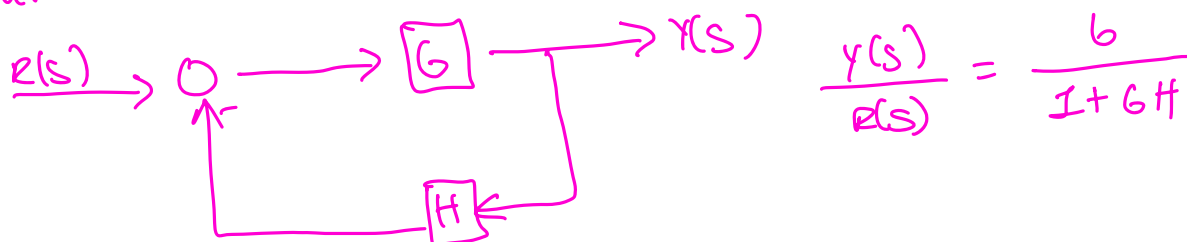
Step 3: Combine cascaded blocks

$$\left(\frac{1}{s+5}\right)\left(\frac{1}{s+10}\right) = \frac{1}{s^2+10s+5s+50} = \frac{1}{s^2+15s+50}$$



Step 4: Eliminate both negative FB signals (1) & (2)

Recall:



$$\frac{Y(s)}{R(s)} = \frac{G}{1+GH}$$

①

$$\frac{\frac{1}{s^2 + 15s + 50}}{1 + \frac{s+5}{s^2 + 15s + 50}} = \frac{1}{s^2 + 15s + 50} \cdot \frac{s^2 + 15s + 50}{s^2 + 16s + 55}$$

$$= \frac{1}{s^2 + 16s + 55} = G, \text{ to be used to find } ②$$

② Using function you get from ① to eliminate f/b

$$\frac{\frac{1}{s^2 + 16s + 55}}{1 + \frac{s+10}{s^2 + 16s + 55}} = \frac{1}{s^2 + 17s + 65}$$

$$\frac{Y(s)}{R(s)} = \frac{1}{s^2 + 17s + 65} \Rightarrow \text{OL form}$$

(B) Loop Gain, $L(s) = (G)(H)$

$$\frac{Y(s)}{R(s)} = \frac{\frac{1}{1}}{\frac{s^2 + 17s}{1} + \frac{65}{1}} = \frac{1}{\frac{1}{6} + H}$$

$$G = \frac{1}{s^2 + 17s}, \quad H = 65$$

$$GH = \frac{65}{s^2 + 17s} = \text{Loop Gain, } L(s)$$

4. Given:

$$V(s) = \frac{400}{s^2 + 8s + 400}$$

Find:

- A. $v(t)$
- B. $v(0)$ using
 - 1. IVT $\Rightarrow V(0) = \lim_{s \rightarrow \infty} sV(s)$
 - 2. evaluating the time domain equation when $t = 0$
- C. $v(\infty)$ using
 - 1. FVT $\Rightarrow V(\infty) = \lim_{s \rightarrow 0} sV(s)$
 - 2. evaluating the time domain equation when $t \rightarrow \infty$
- D. $v(t = 1)$

Complete the Square:

$$s^2 + 8s = -400$$

$$\left(\frac{8}{2}\right)^2 = 16$$

$$s^2 + 8s + 16 = -400 + 16$$

$$(s+4)^2 = -384$$

$$(s+4)^2 + 384$$

From LVT

$$(A) \quad V(s) = \frac{400}{s^2 + 8s + 400} = \frac{400}{(s+4)^2 + 384} \Rightarrow \mathcal{L}^{-1} \left[\frac{\omega_0}{(s+a)^2 + \omega_0^2} \right] = e^{-at} \sin(\omega_0 t)$$

$$a = 4$$

$$\omega_0^2 = 384$$

$$\omega_0 = \sqrt{384} = 8\sqrt{6}$$

$$V(s) = 400 \cdot \frac{1}{(s+4)^2 + 384} \quad \text{need } 8\sqrt{6} \text{ in numerator}$$

$$= \frac{400}{8\sqrt{6}} \cdot \frac{8\sqrt{6}}{(s+4)^2 + 384} = \frac{50}{\sqrt{6}} \cdot \frac{8\sqrt{6}}{(s+4)^2 + 384}$$

$$\frac{50}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{50\sqrt{6}}{6} = \frac{25\sqrt{6}}{3}$$

$$v(t) = \frac{25\sqrt{6}}{3} \mathcal{L}^{-1} \left[\frac{8\sqrt{6}}{(s+4)^2 + 384} \right] = \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6} t)$$

$$v(t) = \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6} t)$$

(B)

1.] $v(0)$ using IVT

$$v(0) = \lim_{s \rightarrow \infty} s V(s) = \lim_{s \rightarrow \infty} s \left[\frac{400}{s^2 + 8s + 400} \right] \bigg|_{s \rightarrow \infty} =$$

$$\frac{400}{1 + \frac{8}{s} + \frac{400}{s^2}} \bigg|_{s \rightarrow \infty} = \frac{0}{1 + 0 + 0} = \frac{0}{1} = \boxed{0}$$

** divide by highest order of s*

2.] evaluate $v(t)$ when $t=0$

$$v(t) = \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6}t)$$

$$v(0) = \frac{25\sqrt{6}}{3} e^{-4(0)} \sin(8\sqrt{6} \cdot 0) = \boxed{0}$$

(C)

1.] $v(\infty)$ using FVT

$$v(\infty) = \lim_{s \rightarrow 0} s V(s) = \lim_{s \rightarrow 0} s \left[\frac{400}{s^2 + 8s + 400} \right] \bigg|_{s \rightarrow 0} = \frac{400s}{s^2 + 8s + 400} \bigg|_{s \rightarrow 0} =$$

$$\frac{0}{400} = \boxed{0}$$

2.] evaluate $v(t)$ when $t \rightarrow \infty$

$$v(t) = \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6}t)$$

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6}t) = \frac{25\sqrt{6}}{3} e^{-4t} \sin(8\sqrt{6}t) \bigg|_{t \rightarrow \infty} = \boxed{0}$$

$$(d) v(t=1) = \frac{25\sqrt{6}}{3} e^{-4(1)} \sin(8\sqrt{6} \cdot 1) = \boxed{0.254}$$

Routh - Hurwitz Stability Criterion

$\Rightarrow a_N s^N + a_{N-1} s^{N-1} + a_{N-2} s^{N-2} + \dots + a_0 = 0$, is the characteristic equation of your transfer function (the denominator in open-loop form $\frac{Y(s)}{U(s)} = \frac{B(s)}{A(s)}$)

OL Form

$$\frac{Y(s)}{U(s)} = \frac{s+3}{s^2+6s+9} \leftarrow \text{Characteristic Equation}$$

$\Rightarrow$ Routh Array set-up

$$\begin{array}{c|cccc} s^N & a_N & a_{N-2} & a_{N-4} & \dots & 0 \\ s^{N-1} & a_{N-1} & a_{N-3} & a_{N-5} & \dots & 0 \end{array} \quad \left. \vphantom{\begin{array}{c|cccc} s^N & a_N & a_{N-2} & a_{N-4} & \dots & 0 \\ s^{N-1} & a_{N-1} & a_{N-3} & a_{N-5} & \dots & 0 \end{array}} \right\} \text{For Step 2}$$

Step 1: Figure out Characteristic Eq

Step 2: Set up first 2 rows of Routh Array

Step 3: Use pivot technique to obtain remaining rows

Step 4: Check for stability [if there are no sign changes
then system is stable]

★ # of sign changes = # of unstable (positive) poles

ex) given: $s^3 + 2s^2 + 4s + 3 = 0$, the characteristic equation

Find: Routh Array

$n = \text{highest degree} = 3$

$\text{Rows} = n + 1 = 4$

s^3	1	4	0
s^2	2	3	0
s^1	$\frac{5}{2}$	0	0
s^0	3	0	0

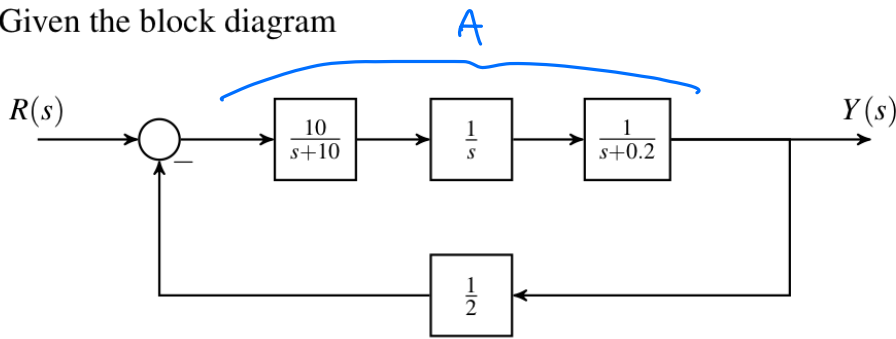
$$\left\{ \begin{aligned} \frac{(2)(4) - (1)(3)}{2} &= \underline{\frac{5}{2}} \\ \frac{2(0) - 1(0)}{2} &= \underline{0} \end{aligned} \right.$$

$$\left\{ \begin{aligned} \frac{\frac{5}{2}(3) - 2(0)}{\frac{5}{2}} &= \frac{\frac{15}{2}}{\frac{5}{2}} = \underline{3} \\ \frac{\frac{5}{2}(0) - 2(0)}{\frac{5}{2}} &= \underline{0} \end{aligned} \right.$$

→ Since there are no sign changes in the 1st column, then system is stable (all roots have negative real parts)

→ # of sign changes = # of unstable (positive) roots of characteristic polynomial in 1st column

5. Given the block diagram

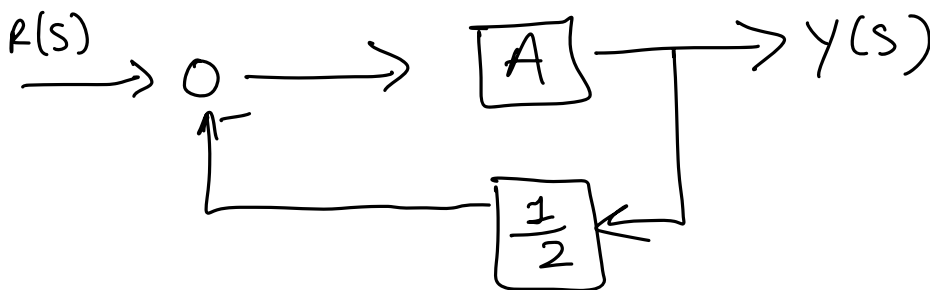


Find:

- The transfer function, $\frac{Y(s)}{R(s)}$
- $y(0)$ using IVT when the input is an i) impulse and ii) a unit step
- $v(\infty)$ using FVT when the input is an i) impulse and ii) a unit step (determine stability with the Routh Array)

(A) step 1: combine cascaded blocks

$$A = \left(\frac{10}{s+10} \right) \left(\frac{1}{s} \right) \left(\frac{1}{s+0.2} \right) = \frac{10}{s(s+10)(s+0.2)}$$



Step 2: Eliminate FB loop

$$\frac{6}{1+6H} = \frac{\frac{10}{s(s+10)(s+0.2)}}{1 + \frac{10}{s(s+10)(s+0.2)} \cdot \frac{1}{2}} = \frac{10}{s(s+10)(s+0.2) + 10}$$

$$= \frac{10}{s(s+10)(s+0.2)} \cdot \frac{2s(s+10)(s+0.2)}{2s(s+10)(s+0.2) + 10}$$

$$= \frac{(2)(10)}{(2)(s)(s+10)(s+0.2) + 10} = \frac{10}{s^3 + 10.2s^2 + 2s + 5}$$

$$\boxed{\frac{Y(s)}{R(s)} = \frac{10}{s^3 + 10.2s^2 + 2s + 5}}$$

$$(B) Y(s) = \frac{10}{s^3 + 10.2s^2 + 2s + 5} R(s)$$

i) IVT when input is impulse

$$r(t) = \delta(t), R(s) = 1$$

$$y(0) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} s \left[\frac{10}{s^3 + 10.2s^2 + 2s + 5} \right] \cdot 1 = \lim_{s \rightarrow \infty} \frac{10s}{s^3 + 10.2s^2 + 2s + 5}$$

$$= \lim_{s \rightarrow \infty} \frac{\frac{10s}{s^3}}{\frac{s^3 + 10.2s^2 + 2s + 5}{s^3}} = \frac{\frac{10}{s^2}}{1 + \frac{10.2}{s} + \frac{2}{s^2} + \frac{5}{s^3}} \bigg|_{s \rightarrow \infty} =$$

$$= \frac{0}{1 + 0 + 0 + 0} = \boxed{0}$$

ii) IVT when input is unit step

$$r(t) = 1, R(s) = \frac{1}{s}$$

$$y(0) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} s \left[\frac{10}{s^3 + 10.2s^2 + 2s + 5} \right] \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{10}{s^3 + 10.2s^2 + 2s + 5} =$$

$$= \frac{\frac{10}{s^3}}{1 + \frac{10 \cdot 2}{s} + \frac{2}{s^2} + \frac{5}{s^3}} \bigg|_{s \rightarrow \infty} = \frac{0}{1+0+0+0} = \frac{0}{1} = \boxed{0}$$

(c) i) FVT when input is impulse

$$r(t) = \delta(t), \quad R(s) = 1$$

$$y(\infty) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \left[\frac{10}{s^3 + 10 \cdot 2s^2 + 2s + 5} \right] \cdot \frac{1}{s} = \frac{10s}{s^3 + 10 \cdot 2s^2 + 2s + 5} \bigg|_{s \rightarrow 0} =$$

$$= \frac{0}{0+0+0+5} = \frac{0}{5} = \boxed{0}$$

ii) FVT when input is unit step

$$r(t) = 1, \quad R(s) = \frac{1}{s}$$

$$y(\infty) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \left[\frac{10}{s^3 + 10 \cdot 2s^2 + 2s + 5} \right] \cdot \frac{1}{s} = \frac{10}{s^3 + 10 \cdot 2s^2 + 2s + 5} \bigg|_{s \rightarrow 0} =$$

$$= \frac{10}{0+0+0+5} = \frac{10}{5} = \boxed{2}$$

Routh Array

$$s^3 + 10 \cdot 2s^2 + 2s + 5$$

s^3	1	2	0
s^2	10.2	5	0
s^1	1.51	0	0
s^0	5	0	0

10.2 is pivot

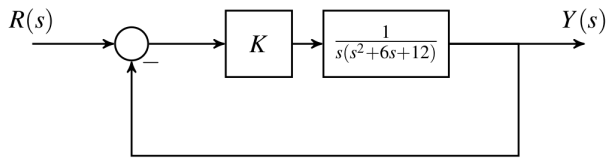
$$\left[\begin{array}{l} \frac{10 \cdot 2(2) - 1(5)}{10 \cdot 2} = 1.51 \\ \frac{0 - 0}{10 \cdot 2} = 0 \end{array} \right.$$

1.51 is pivot

$$\left[\begin{array}{l} \frac{7.55 - 0}{1.51} = 5 \\ \frac{0 - 0}{1.51} = 0 \end{array} \right.$$

No sign changes,
so stable

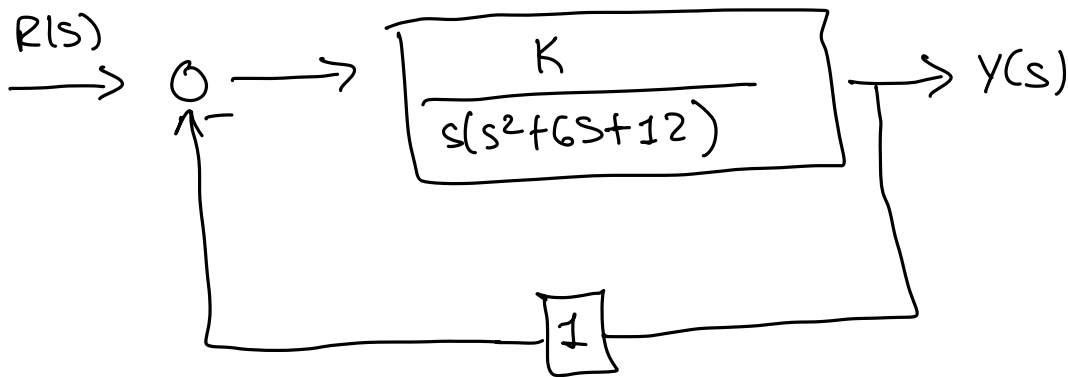
6. Given the block diagram



Find:

- A. The transfer function, $\frac{Y(s)}{R(s)}$
- B. The loop-gain

(A) Step 1: combine cascaded blocks



Step 2: Eliminate Negative FB signal

$$\frac{G}{1+GH} = \frac{\frac{K}{s(s^2+6s+12)}}{1 + \frac{K}{s(s^2+6s+12)}} = \frac{K}{s(s^2+6s+12)} \cdot \frac{s(s^2+6s+12)}{s(s^2+6s+12)+K}$$

$$= \frac{K}{s(s^2+6s+12)+K} = \frac{K}{s^3+6s^2+12s+K}$$

$$\boxed{\frac{Y(s)}{R(s)} = \frac{K}{s^3+6s^2+12s+K}}$$

(B) Find Loop Gain, $L(s) = GH$

$$\frac{\frac{K}{K}}{\frac{s^3 + 6s^2 + 12s + K}{K}} = \frac{1}{\frac{s^3 + 6s^2 + 12s}{K} + \frac{K}{K}} = \frac{1}{\frac{1}{G} + H}$$

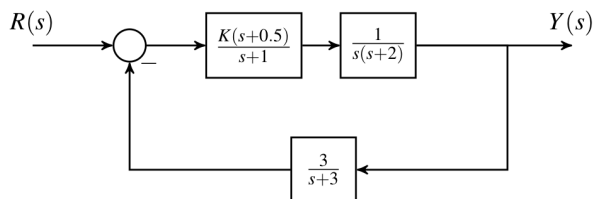
$$G = \frac{K}{s^3 + 6s^2 + 12s}$$

$$H = 1$$

loop Gain = $L(s) = GH =$

$$\frac{K}{s^3 + 6s^2 + 12s}$$

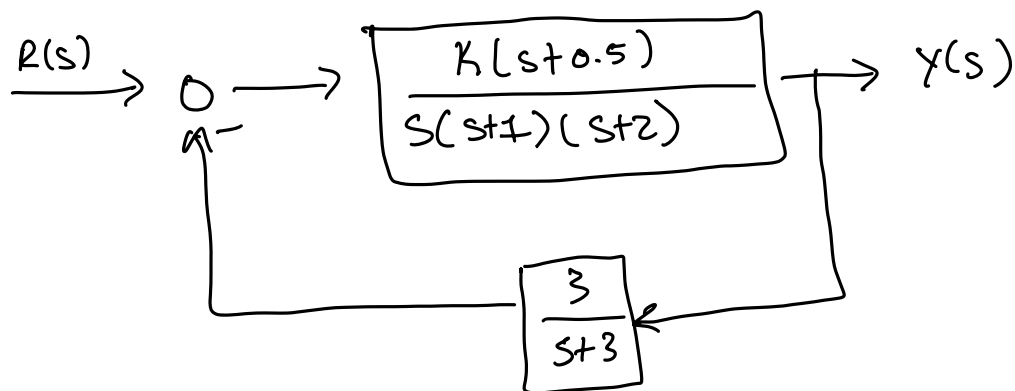
7. Given the block diagram



Find:

- The transfer function, $\frac{Y(s)}{R(s)}$
- The loop-gain

(A) Step 1: Combine Cascaded Blocks



Step 2: Eliminate Negative FB Signal

$$\frac{G}{1+GH} = \frac{\frac{K(s+0.5)}{s(s+1)(s+2)}}{1 + \frac{3K(s+0.5)}{s(s+1)(s+2)(s+3)}} = \frac{K(s+0.5)}{\cancel{s(s+1)(s+2)}} \cdot \frac{\cancel{s(s+1)(s+2)}(s+3)}{s(s+1)(s+2)(s+3) + 3K(s+0.5)} =$$

$$= \frac{K(s+0.5)(s+3)}{s(s+1)(s+2)(s+3) + 3K(s+0.5)} = \frac{Ks^2 + 3.5Ks + 1.5K}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks + 1.5K}$$

$$\frac{Y(s)}{R(s)} = \frac{Ks^2 + 3.5Ks + 1.5K}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks + 1.5K}$$

$$(B) \frac{Ks^2 + 3.5Ks + 1.5K}{Ks^2 + 3.5Ks + 1.5K} = \frac{1}{\frac{s^4 + 6s^3 + 11s^2 + 6s + 3Ks}{Ks^2 + 3.5Ks + 1.5K} + \frac{1.5}{s^2 + 3.5s + 1.5}}$$

$$\frac{1}{\frac{1}{G} + H} \rightarrow G = \frac{Ks^2 + 3.5Ks + 1.5K}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks} \quad H = \frac{1.5}{s^2 + 3.5s + 1.5}$$

$$\text{loop gain} = GH = \left(\frac{Ks^2 + 3.5Ks + 1.5K}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks} \right) \left(\frac{1.5}{s^2 + 3.5s + 1.5} \right) =$$

$$= \left(\frac{K(s^2 + 3.5s + 1.5)}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks} \right) \left(\frac{1.5}{s^2 + 3.5s + 1.5} \right)$$

$$\text{loop gain} = \frac{1.5K}{s^4 + 6s^3 + 11s^2 + 6s + 3Ks}$$

8. Given: the characteristic polynomial:

$$A(s) = s^3 + 8s^2 + 29s + 52$$

Find:

A. The number of unstable roots

$$A(s) = s^3 + 8s^2 + 29s + 52$$

s^3	1	29	0	$\frac{8(29) - 52(1)}{8} = 22.5$ $\frac{0 - 0}{8} = 0$
s^2	8	52	0	
s^1	22.5	0	0	$\frac{22.5(52) - 8(0)}{22.5} = 52$ $\frac{0 - 0}{22.5} = 0$
s^0	52	0	0	

No sign changes, so
no unstable roots

9. Given: the characteristic polynomial:

$$A(s) = s^3 - 3s - 52$$

Find:

A. The number of unstable roots

$$A(s) = s^3 - 0s^2 - 3s - 52$$

s^3	1	-3	0	$\frac{0 - (-52)}{0} = 0$ $\frac{0 - 0}{0} = 0$
s^2	0	-52	0	
s^1	0	0	0	
s^0	0	0	0	

No sign changes, so
no unstable roots

10. Given: the characteristic polynomial:

$$A(s) = s^3 + 4s^2 + 13s$$

Find:

$$A(s) = s^3 + 4s^2 + 13s + 0s^0$$

A. The number of unstable roots

$$\begin{array}{l|rrr} s^3 & 1 & 13 & 0 \\ s^2 & 4 & 0 & 0 \\ s^1 & 13 & 0 & 0 \\ s^0 & 0 & 0 & 0 \end{array}$$

$$\frac{4(13) - 1(0)}{4} = 13$$

$$\frac{0 - 0}{4} = 0$$

No sign changes, so
no unstable roots